Machine Learning for Traditional Indian Crop Prediction

Case Study with Indian Knowledge Systems (IKS)

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Introduction

- **Agriculture:** The backbone of the Indian economy for centuries.
- **Traditional Knowledge:** Farmers rely on experiential knowledge (soil type, rainfall patterns, seasonal cycles).
- **The Challenge:** Traditional practices can lack precision against rapidly changing environmental conditions.
- **The Solution:** Machine Learning (ML) provides a data-driven approach to enhance consistency and accuracy.

Problem Statement

Objective: Develop an ML model to predict the most suitable crop based on environmental and soil conditions.

Inputs:

- Soil Type (Black, Red, etc.)
- Rainfall (Numeric/Levels)
- Temperature (°C)
- Season (Kharif, Rabi)

Output:

- Recommended Crop (Rice, Wheat, Millet, Cotton, etc.)

Objectives

1. Design a predictive system for automated crop recommendation.
2. Bridge the gap between **Traditional Agricultural Knowledge** and **Modern AI techniques**.
3. Analyze and compare the performance of different Machine Learning algorithms.

Dataset Description

Feature	**Description**	**Examples**
Soil Type	Regional soil variant	Black, Red, Sandy
Rainfall	Precipitation levels	Low, Medium, High
Temperature	Average temp ($^\circ$C)	20–35
Season	Growing cycle	Kharif, Rabi
Crop	Target label	Rice, Wheat, Cotton

Table: Agricultural parameters used in the study.

Methodology

1. Data Preprocessing

- Categorical variables (Soil, Rainfall, Season) encoded via **One-Hot Encoding**.
- Numeric features (Temperature) normalized for the model.
- Data Split: 80% Training / 20% Testing.

2. Model Selection

- **Decision Tree:** Interpretative model based on feature splits.
- **Random Forest:** Ensemble method to reduce overfitting.

Results & Applications

Example Prediction:

- *Input:* Black Soil, High Rainfall, $30°$C, Kharif.
- *Output:* **Rice**.

Applications:

- Real-time crop selection assistance for farmers.
- Risk reduction regarding crop failure.
- Integration into mobile apps for remote access.

Advantages & Limitations

Advantages:

- Scalable to different regions.
- Data-driven consistency.
- Merges IKS with AI.

Limitations:

- Dependent on data quality.
- Ignores sudden climate shifts.
- Needs constant updates.

Conclusion

- This project demonstrates how ML can modernize Indian agricultural practices.
- By leveraging environmental data, farmers can make informed decisions.
- Future work could include real-time climate API integration and soil nutrient analysis for higher precision.

Thank You!